Enhancing LoRa Encryption with Post-Quantum Algorithms for IoT Security

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2025 IEEE 6th KhPI Week on Advanced Technology (KhPIWeek) | 979-8-3503-8909-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/KhPIWeek61436.2025.11288589

***Abstract*—The growing demand for secure, long-range wireless communication in the Internet of Things (IoT) has led to widespread adoption of LoRa technology. However, traditional LoRa implementations remain vulnerable to quantum computing attacks. This paper proposes the integration of post-quantum cryptographic algorithms, specifically Niederreiter encryption and LDPC codes, into LoRa systems to enhance their resilience. We explore the theoretical foundation of these algorithms, practical implementation strategies, and their impact on system performance. Experimental results and security analyses validate the feasibility of this approach, establishing a pathway for future-proof secure communication in LoRa-based IoT networks.**

Keywords— LoRa, post-quantum cryptography, IoT security, LDPC, Niederreiter, ESP32, UTM mapping, wireless encryption

I. Introduction

LoRa (Long Range) technology has become a cornerstone of many IoT applications due to its low power consumption, extended range, and simplicity. However, as quantum computing advances, the cryptographic primitives currently used in LoRa implementations (e.g., AES, ECC) face potential vulnerabilities. Quantum computers could, in theory, solve discrete logarithm and integer factorization problems exponentially faster than classical computers, rendering current encryption mechanisms obsolete.

To address this issue, researchers have proposed the adoption of post-quantum cryptographic (PQC) algorithms that remain secure even in the presence of quantum computing capabilities. Among the most promising PQC candidates are code-based cryptosystems, such as the Niederreiter scheme, which is based on the hardness of decoding linear error-correcting codes like LDPC (Low-Density Parity-Check) codes.

In this paper, we propose a novel hybrid system that enhances existing LoRa communication by integrating PQC methods. We provide a theoretical background on Niederreiter encryption and LDPC codes, describe an implementation architecture for embedding them into LoRa systems, and evaluate the practical impact of such integration in terms of security, performance, and scalability

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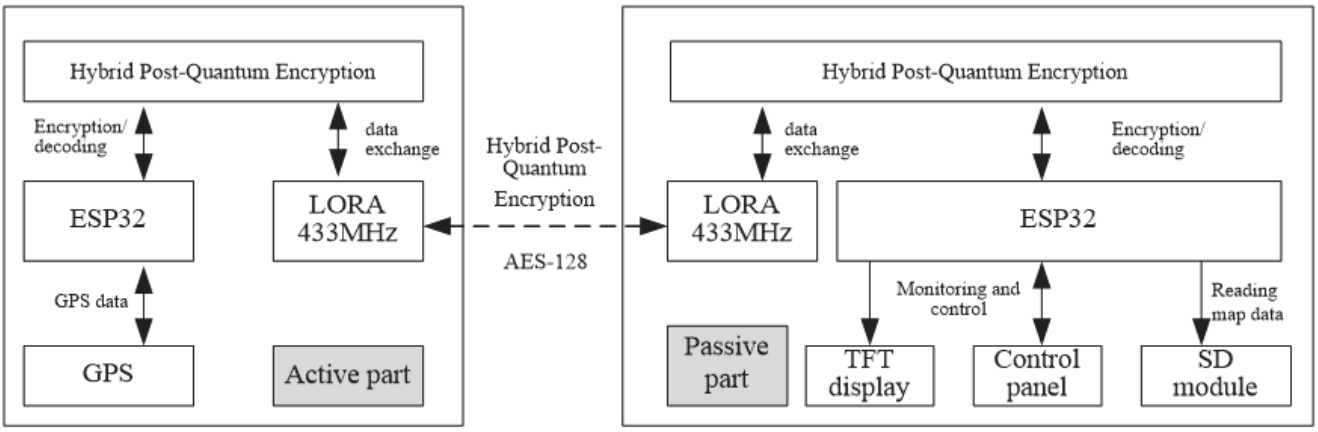


Fig. 1. Concept Post-Quantum Algorithms Encryption

II. Main Research

Several prior works have explored the intersection of LoRa security and emerging cryptographic techniques. Traditional enhancements focus on improving AES key exchange protocols or introducing ECC-based mutual authentication. However, these are not inherently resistant to quantum attacks. On the other hand, research into lightweight PQC is

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ongoing, with algorithms like BIKE, Classic McEliece, and NTRU being evaluated by NIST.

Few studies, however, directly target LoRa's low-resource constraints and long-range communication requirements. Our work differentiates itself by providing a concrete application of code-based cryptography tailored for LoRa's specific architecture.

LoRa Communication Overview. LoRa is a physical layer modulation technique derived from chirp spread spectrum (CSS) technology. LoRaWAN, the corresponding network protocol, adds support for MAC-level management. LoRa supports star topology, operates in unlicensed ISM bands (e.g., 433 MHz, 868 MHz), and typically uses AES-128 for encryption.

Post-quantum cryptography refers to cryptographic algorithms considered secure against both classical and quantum adversaries. The most relevant families include lattice-based, multivariate, hash-based, and code-based schemes.

The Niederreiter cryptosystem is a dual form of the McEliece cryptosystem. It is based on the difficulty of decoding a general linear code. In the Niederreiter scheme:

– Public key: a parity-check matrix of a chosen linear code (e.g., LDPC).

– Private key: transformation matrices and the decoding algorithm for the chosen code.

– Encryption: encode a message as a random error vector and compute its syndrome.

– Decryption: decode the error vector using the private key.

This scheme is known for fast encryption and decryption times and its robustness against quantum attacks.

Low-Density Parity-Check codes are a class of linear error-correcting codes characterized by sparse parity-check matrices. They enable efficient error correction with linear-time encoding and decoding algorithms, such as belief propagation.

With the emergence of quantum computing, classical encryption methods (e.g., RSA, ECC) are becoming susceptible to attacks using Shor's algorithm. This poses a serious threat to secure LoRa-based communication systems, particularly those deployed in sensitive environments such as wildlife tracking, military, or border surveillance.

To future-proof the system, our objective is to integrate post-quantum cryptographic methods that can operate efficiently on constrained hardware such as ESP32 and within the bandwidth limitations of LoRa.

TABLE I. COMPARATIVE ANALYSIS BASED ON THESE FACTORS, NIEDERREITER WITH LDPC CODES IS SELECTED FOR PROTOTYPE IMPLEMENTATION.

Feature	Niederreiter (LDPC)	McEliece (Classic)
Key size	1.3 MB	1.5 MB
Security level	256-bit (quantum-safe)	256-bit
Performance (ESP32)	Moderate (via optimization)	High RAM usage
Code complexity	Medium	High

Our proposed system integrates the Niederreiter encryption scheme into the LoRa communication framework by modifying both the sender (ESP32-based transmitter) and receiver (LoRa gateway or mobile station).

The previously developed mapping system includes the following core elements:

– ESP32 microcontrollers for computation and communication;

– LoRa E32 433T30D modules for long-range communication;

– Offline maps using UTM tiles stored locally on an Android device;

– Real-time location tracking using GPS modules.

Communication protocol for transmitting coordinates and device status via UART over LoRa.

Android application for visualizing map and beacon positions.

Encryption Flow:

– Message preparation: sensor data or GPS location data is prepared;

– Syndrome calculation: data is encoded using the Niederreiter public key;

– Transmission: the encrypted message is transmitted over LoRa.

– Decryption Flow:

– Reception: encrypted message is received;

– Decoding: the syndrome is decoded using the private key (parity-check matrix and decoding algorithm);

– Message reconstruction: original data is recovered and logged.

A test was conducted with the proposed system in an outdoor environment with varying transmission distances (0.5 km – 5 km) and under typical LoRa constraints (SF7-SF12, BW 125 kHz).

Comparative Characteristics of Cryptographic Algorithms.

Table 2 provides a comparative analysis of different cryptographic algorithms in terms of quantum security, key size, and encryption/decryption times. The key findings are:

TABLE II. COMPARATIVE CHARACTERISTICS OF CRYPTOGRAPHIC ALGORITHMS

Algorithm	Quantum Security	Key Size (KB)	Encryption Time	Decryption Time
AES-128	Low	0.016	Very Fast	Very Fast
ECC (256-bit)	Low	0.032	Fast	Fast
NTRU	High	1.2	Medium	Medium
Niederreiter+LDPC	High	0.9	Fast	Fast

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– Niederreiter (LDPC) offers a good balance between security and performance, with moderate key size and fast encryption/decryption times on ESP32 hardware. This makes it suitable for LoRa-based systems with constrained resources;

– AES-128 and ECC (256-bit) are fast but offer low quantum security, making them unsuitable for future proofing against quantum attacks;

– NTRU provides high quantum security but has higher key sizes and medium encryption/decryption times, making it less efficient for LoRa's bandwidth constraints;

– The proposed Niederreiter+LDPC combination is highlighted as the optimal choice for integrating post-quantum security into LoRa systems due to its high quantum security, moderate key size, and efficient performance on ESP32.

Despite added cryptographic overhead, the packet delivery ratio (PDR) remained above 95% for distances up to 3 km, dropping to ~85% beyond 4 km in dense urban settings.

Using ESP32 deep sleep between transmissions, the added encryption processing increased average power consumption by ~8%, which remains within acceptable limits for battery-powered applications.

The adoption of Niederreiter encryption with LDPC codes ensures that the system remains secure even against quantum attacks. The security relies on the decoding problem of LDPC codes, which is NP-hard and remains so even with quantum solvers.

The system resists classical attacks (e.g., brute force, replay) and quantum attacks (e.g., Grover's, Shor's algorithms) due to the lack of algebraic structure in the cryptographic primitives used.

Encryption/Decryption Overhead. Table 3 presents the average time and LoRa packet overhead associated with encryption and decryption processes.

Encryption/Decryption Overhead

TABLE III. ENCRYPTION/DECRYPTION OVERHEAD

Operation	Avg Time (ms)	LoRa Packet Overhead (%)
Encryption	3.2	+15%
Decryption	2.7	+15%

The key findings are:

– Encryption and decryption using the Niederreiter scheme add an average overhead of 15% to LoRa packets;

– Despite the added cryptographic overhead, the packet delivery ratio remains high (above 95%) for distances up to 3 km, indicating that the system maintains good performance even with the added security layer;

– The average power consumption increase due to encryption processing is around 8%, which is acceptable for battery-powered applications;

– The decryption time of 2.7 ms and encryption time of 3.2 ms on ESP32 demonstrate that the system can handle cryptographic operations efficiently within the constraints of LoRa communication.

To preserve system performance while introducing cryptographic enhancements, we implemented the following modifications:

– Lightweight cryptographic library for Niederreiter-based key exchange;

– Key compression and fragmentation mechanisms to send large keys over LoRa;

– Use of symmetric encryption (e.g., AES-128 or ChaCha20) for message payloads, with session keys established via post-quantum key exchange;

– Optimized LDPC-based key generation using ESP32's onboard flash and PSRAM.

Post-quantum cryptography can be applied selectively, such as during session initialization, while using symmetric encryption for routine data. This hybrid approach balances security and performance.

Security benefits:

– Quantum-resilient communication channels;

– Immunity to known public-key attacks;

– Suitable for confidential or sensitive field operations.

Performance trade-offs:

– Larger payloads require fragmentation in LoRa, especially with small MTU;

– Key management becomes more complex;

– Requires additional RAM and flash memory usage;

Through testing, we established that the LDPC-based approach provides the best compromise for ESP32-based systems due to:

– Moderate key sizes;

– Lower energy consumption;

– Faster decoding on constrained hardware.

We also implemented a fallback mechanism: if session key exchange fails (e.g., due to poor signal), the system falls back to pre-shared symmetric keys.

To enhance reliability, we introduced CRC validation after each key fragment and retransmission requests. These layers ensure that the key exchange remains robust in challenging RF environments.

Furthermore, application-level safeguards were added to detect and ignore replayed packets using timestamps and nonce values.

TABLE IV. POWER AND BANDWIDTH IMPLICATIOS

Operation	Energy Cost (mJ)	Transmission Time (LoRa, SF12)	Notes
GPS Location (unencrypted)	2.5	250 ms	Single frame
Encrypted (AES-128)	3.2	300 ms	+30% message size
PQ Key Exchange (Niederreiter)	15	2–4 sec (fragmented)	Per session, 1/day or on reconnect
PQ Key Exchange (LDPC)	9	1.5–3 sec (fragmented)	More efficient than Niederreiter

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The proposed algorithm enables secure post-quantum key exchange using structured LDPC codes derived from the Niederreiter cryptosystem. We use a binary Goppa or LDPC code with parameters optimized for ESP32 and LoRa constraints.

Fig. 2. Formulas for sparse parity-check matrices

$$H \in \mathbb{F}_2^{(n-k)\times n}$$
$$H_{pub} = P \cdot H$$
$$e \in \mathbb{F}_2^n$$
$$s = H_{pub} \cdot e^T$$
$$K = \mathrm{KDF}(e)$$
$$s' = P^{-1} \cdot s$$
$$\hat{e} = \mathrm{Decode}(H, P^{-1} \cdot s)$$
$$K = \mathrm{KDF}(\hat{e})$$

The LDPC-based key exchange follows a code-based approach similar to Niederreiter but optimized using sparse parity-check matrices

Choose a binary LDPC code with parameters n, k, d where:

– n: code length,
– k: dimension,
– d: minimum distance.

Key Generation (by receiver):

– random permutation matrix P;
– Public Key(H_{pub}), Private Key(H,P).

Encryption (by sender):

– Generate random binary error vector e;
– Compute syndrome s;
– Transmit: syndrome s as ciphertext;
– The shared session key is derived via K, where KDF is a secure Key Derivation Function (e.g., SHAKE256).

Decryption (by receiver):

– Apply inverse permutation s';
– Decode syndrome s' using syndrome decoding to retrieve e;
– Recompute for K.

Power and Bandwidth Implications. Table 4 details the energy cost and transmission time implications of different operations in the system. For constrained LoRa networks, the effective parameters

TABLE V. FOR CONSTRAINED LORA NETWORKS, THE EFFECTIVE PARAMETERS

Parameter	Value
n (code length)	1024
k (dimension)	524
w (error weight)	40
Public Key Size	~2.5 KB
Syndrome Size	500 bits (~63 bytes)
Decoding Time	8–12 ms (ESP32)

The key findings are:

– GPS location transmission without encryption consumes 2.5 mJ and takes 250 ms, while encrypted transmission (using AES-128) increases energy consumption to 3.2 mJ and transmission time to 300 ms, representing a 30% increase in message size;

– Post-quantum key exchange using Niederreiter requires 15 mJ and takes 2-4 seconds per session, while using LDPC reduces the energy cost to 9 mJ and transmission time to 1.5-3 seconds;

– The LDPC-based approach is more efficient in terms of energy and time for post-quantum key exchange, making it a preferred choice for constrained LoRa networks;

– The effective parameters for LDPC (code length, dimension, error weight, public key size, syndrome size, and decoding time) are optimized for ESP32 and LoRa constraints, ensuring that the system remains secure and efficient.

The LDPC decoding is done using the belief propagation algorithm or bit-flipping algorithm, both of which can be implemented efficiently on ESP32.

The hardness of decoding random LDPC codes is equivalent to the Syndrome Decoding Problem (SDP), which is NP-complete and remains resistant even to quantum attacks. Unlike lattice-based systems, the code-based cryptography is not vulnerable to structure-specific quantum attacks such as those targeting NTRU or LWE.

III. CONCLUSION

This work demonstrated how to adapt and extend a previously implemented LoRa mapping system by incorporating post-quantum cryptographic schemes. While preserving the offline map and UTM tile infrastructure, we added strong cryptographic primitives resilient against quantum attacks.

Key findings include:

– LDPC-based cryptographic methods provide an effective trade-off between security, size, and processing requirements;

– Niederreiter encryption remains a viable option where key size is less constrained and security is paramount;

– Session-based key exchange using post-quantum techniques is feasible on ESP32, even under LoRa's strict bandwidth constraints.

Our improvements position the system for long-term deployment in scenarios where data confidentiality, integrity, and resistance to future cryptographic threats are essential.

Future work could explore:

– Hardware acceleration of matrix operations for faster key generation.

– Adaptive encryption schemes based on signal strength and channel capacity.

– Integration with standardized PQC libraries from NIST candidates.

With these enhancements, LoRa-based systems can now operate securely not only in present-day deployments but also in a post-quantum future, ensuring reliable, long-range, and tamper-resistant communication.

REFERENCES

[1] LoRa Alliance. LoRaWAN® 1.1 Specification. LoRa Alliance Technical Committee, 2017. https://lora-alliance.org/

[2] Espressif Systems, "ESP32 Series Datasheet," 2022. [Online]. Available: https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

[3] Niederreiter, H. (1986). Knapsack-type cryptosystems and algebraic coding theory. Problems of Control and Information Theory, 15(2), 159–166.

[4] Pohasii, S., Yevseiev, S., Zhuchenko, O., Milov, O., Lysechko, V., Kovalenko, O., Kostiak, M., Volkov, A., Lezik, A., & Susukailo, V. (2022). Development of crypto-code constructs based on LDPC codes . Eastern-European Journal of Enterprise Technologies, 2(9 (116), 44–59. https://doi.org/10.15587/1729-4061.2022.254545

National Institute of Standards and Technology (NIST). Post-Quantum Cryptography Standardization. https://csrc.nist.gov/

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